MIL-S-2244A(EL)
7 MAR 1967

SUPERSEDING
MIL-S-2244
30 June 1950

MILITARY SPECIFICATION

SLEEVES, WIRE SPLICING, SINGLE TUBE,

GENERAL SPECIFICATION FOR

1. SCOPE

1.1 Scope - This specification covers the general requirements for sleeves, wire splicing, single tube, for use in making splices in copper, copper-clad, and steel wires.

1.2 Classification -

1.2.1 Types - Single tube wire splicing sleeves shall be of the sleeve material and the commercial designations in general use by the manufacturers of these sleeves, or equals, as follows:

| | |
|---|---|
| C-080C | Copper |
| C-104C | Copper |
| C-104Q | Copper |
| S-109Q | Galvanized steel |
| S-134Q | Galvanized steel |

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitations for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-Z-325 - Zinc Coating, Electrodeposited, Requirements for

WW-T-797 - Tube, Copper (Seamless)

PPP-B-636 - Box, Fiberboard

PPP-B-640 - Box, Fiberboard, Corrugated, Triple-Wall

PPP-S-760 - Strapping, Nonmetallic and Connectors

PPP-T-97 - Tape, Pressure-Sensitive Adhesive, Filament Reinforced

FSC 5975

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MIL-P-116 - Preservation, Methods of

MIL-Z-17671 - Zinc-Coating, Hot Dip Galvanizing

MIL-B-43291 - Boxes, Fiberboard, Corrugated, Double-Wall, Weather Resistant

MIL-C-45662 - Calibration of Standards

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-130 - Identification Marking of U. S. Military Property.

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A90 Standard Method of Test for Weight of Coating on Zinc-Coated (Galvanized) Iron or Steel Articles.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, 3 Pennsylvania).

3. REQUIREMENTS

3.1 Materials - Materials used in the manufacture of single tube wire splicing sleeves shall be as specified herein. Materials which are not specifically designated shall be of the best quality and suitable for purpose intended. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.



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3.1.2 Metals - Single tube wire splicing sleeve material shall be either copper tubing or steel as specified (see 1.2.1).

3.1.2.1 Copper tubing - Copper tubing shall be in accordance with specification WW-T-797.

3.1.2.2 Steel - Steel shall be AISI 1010C, SAE 1 temper steel.

3.1.3 Finishes -

3.1.3.1 Zinc coating - The outer cylindrical surfaces and end surfaces of single tube wire splicing steel sleeves shall be covered with a smooth, continuous, adherent and uniform zinc coating. The coating may be applied either by hot-dip galvanizing in accordance with Specification MIL-Z-17871 or by electrodepositing in accordance with QQ-Z-325. Special care should be taken to insure that no zinc is deposited in the bore of the sleeves.

3.1.3.2 Bore abrasive coating - To improve the gripping strength of the single tube wire splicing sleeves, the bores shall receive a thin coating throughout their entire length consisting of, or containing, an abrasive material. The bores of all single tube wire splicing sleeves shall be clean and bright before the application of the coating of abrasive. This coating may consist of a well-distributed hard metal alloy, such as nichrome, sprayed in a molten state on the bore of the single tube wire splicing sleeve; a mixture of emery granules with a thin, durable, clear nitro-cellulose lacquer binder; or any other conductive abrasive coating acceptable to the contracting officer. When lacquer is used it shall be applied in such a manner that there will be no appreciable flow in the bores as indicated by the presence of beads, drops, or uneven coating on the surface of the bores.

3.1.3.2.1 Emery (when applicable) - If used as the abrasive material, emery shall have the following distribution of grain size for one hundred gram samples simultaneously agitated for five minutes on a Rotap machine at a speed of 290 rpm and 156 taps per minute employing the following U. S. Standard sieve series: All material shall pass through a No. 120 sieve; not more than fifteen percent shall be retained on a No. 170 sieve; at least sixty-five percent of the material passing the No. 170 sieve shall be retained on Nos. 200, 230, or 270 sieves. Not more than three percent shall pass through a No. 325 sieve.

3.2 Design and construction - Single tube wire splicing sleeves covered by this specification shall be of the design, construction, and physical dimensions specified in figure 1.

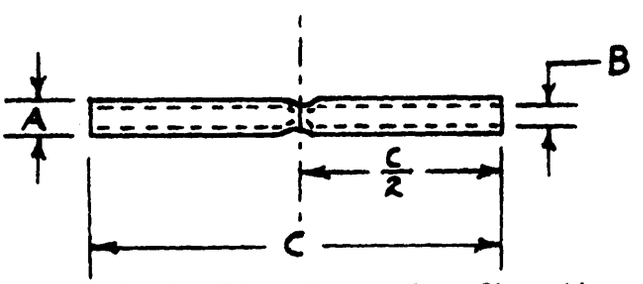



FIGURE 1. Dimensions and configuration



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| TYPES | DIMENSIONS | | | | |
|---|---|---|---|---|---|
| | A (After Zinc coating, when applicable) | | B (After Abrasive coating) | | C (Min.) |
| | Max. | Min. | Go-Gage 1/ +0 -.0005 | No-go-Gage +.0005 - 0 | |
| C-080C | 0.202 (5.13 mm) | 0.192 (4.88 mm) | 0.0850 (2.159 mm) | 0.0920 (2.336 mm) | 1.375 (34.93 mm) |
| C-104C | 0.202 (5.13 mm) | 0.192 (4.88 mm) | 0.1080 (2.743 mm) | 0.1150 (2.921 mm) | 1.435 (36.45 mm) |
| C-104Q | 0.251 (6.38 mm) | 0.244 (6.20 mm) | 0.1080 (2.743 mm) | 0.1150 (2.921 mm) | 2.313 (58.75 mm) |
| S-109Q | 0.249 (6.32 mm) | 0.245 (6.22 mm) | 0.1140 (2.895 mm) | 0.1200 (3.048 mm) | 2.500 (63.50 mm) |
| S-134Q | 0.249 (6.32 mm) | 0.245 (6.22 mm) | 0.1400 (3.556 mm) | 0.1460 (3.708 mm) | 2.313 (58.75 mm) |

1/ The Go-Gages specified shall have a length equivalent to one and one-half times the nominal diameter of the wire with which the sleeve is to be employed and the sleeves shall have a sufficiently large bore to permit insertion of the full length of the gage.

NOTES:

1. All dimensions in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. Drawing depicts only general design characteristics and is not intended to limit specific manufacturing processes.

FIGURE 1. Dimensions and configuration. (Con't)

3.3 Indentation - A rounded indentation, transverse or circumferential, which does not cut the metal of the single tube wire splicing sleeve, shall constrict the bore of the single tube wire splicing sleeve midway of its length. In galvanized steel wire splicing sleeves which have their bores formed by drilling, the constriction may be in the form of a ring, of triangular cross-section, located at the mid-point of the bore. The form and position of the constriction and the uniformity and straightness of the bore shall be such as to permit the entrance of the gage specified in table 1 for the required distance. The constriction shall not close the bore but shall prevent the passage of a cylindrical gage having a diameter 0.003 inch less than the nominal diameter of the wire for which the single tube wire splicing sleeve is intended.



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TABLE I. Gage diameter and insertion depth (in inches).

| Type | Diameter of Gage (± .0005) | Depth of Gage Insertion |
|---|---|---|
| C-080C | 0.0830 (2.108 mm) 1/ | 0.625 (15.88 mm) |
| C-104C | 0.1060 (2.692 mm) | 0.656 (16.67 mm) |
| C-104Q | 0.1060 (2.692 mm) | 1.094 (27.79 mm) |
| S-109Q | 0.1130 (2.870 mm) | 1.187 (30.16 mm) |
| S-134Q | 0.1380 (3.505 mm) | 1.094 (27.79 mm) |

1/ Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.

3.4 Weight of Zinc Coating - When tested as specified in 4.5.2 the weight of zinc coating on the outer surfaces, of the galvanized steel single tube wire splicing sleeves, including end surfaces, shall be not less than 150 milligrams per square inch of surface.

3.5 Adherence of Zinc Coating - When tested as specified in 4.5.3 zinc coating shall not flake or loosen sufficiently to permit removal of the zinc by rubbing with the fingers nor shall the galvanized steel single tube wire splicing sleeves fracture during this test.

3.6 Gripping strength - When tested as specified in 4.5.4 the single tube wire splicing sleeve shall not break while being attached by the specified rolling or pressing tools (see 4.5.4) nor slip, after attachment, upon the wire with which it is intended to be used.

3.7 Marking - Single tube wire splicing sleeves shall be marked in accordance with Standard MIL-STD-130, with the type and manufacturers name or symbol in any convenient external location.

3.8 Workmanship - All single tube wire splicing sleeves shall be uniform in quality and shall be free from splits, scales or any defect that affects life, serviceability, or appearance. Single tube wire splicing sleeve bores shall be as straight and concentric as possible.



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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Inspection - The examination and testing of single tube wire splicing sleeves shall be classified as follows:

(a) Component - Materials Inspection (see 4.3).

(b) Quality Conformance Inspection (see 4.4).

(1) Inspection of product for delivery (see 4.4.1).

(2) Inspection of preparation for delivery (see 4.4.2).

4.2 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2.1 Test equipment and inspection facilities - Test equipment and inspection facilities shall be of sufficient accuracy, quality and quantity to permit performance of the required inspection. The supplier shall establish calibration of test equipment to the satisfaction of the Government. Calibration of the standards which control the accuracy of the inspection equipment shall comply with the requirements of MIL-C-45662.

4.2.2 Test and inspection conditions - Unless otherwise specified herein, test and inspection conditions shall be in accordance with paragraph 2.2 of the general requirements of MIL-STD-202.

4.3 Component--materials inspection - Component--materials inspection shall consist of verification that the component materials, listed in Table II, used in fabricating the single tube wire splicing sleeves, are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

TABLE II. Component--materials inspection

| Component--Material | Requirement paragraph | Applicable specification |
|---|---|---|
| Metals: | | |
| Copper tubing | 3.1.2.1 | WW-T-797 |
| Steel | 3.1.2.2 | |
| Emery (when applicable) | 3.1.3.2.1 | |
| Finishes: | | |
| Zinc coating | 3.1.3.1 | MIL-Z-17871 or QQ-Z-325 |
| Bore abrasive coating | 3.1.3.2 | |



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4.4 Quality conformance inspection -

4.4.1 Inspection of product for delivery - Inspection of product for delivery shall consist of groups A and B for single tube wire splicing sleeves.

4.4.1.1 Inspection lot - An inspection lot, as far as practicable, shall consist of all the single tube wire splicing sleeves, of the same designation, produced under essentially the same conditions, and offered for inspection at one time.

4.4.1.2 Resubmitted lots - If an inspection lot is rejected, the supplier may replace it with a new lot, rework it to correct the defects, or screen out the defective units, and submit it again for inspection. Resubmitted lots shall be kept separate from new lots and shall be clearly identified as resubmitted lots. Resubmitted lots shall be inspected, using tightened inspection.

4.4.1.3 Group A inspection - Group A inspection shall consist of examinations specified in Table III, and shall be made on the same set of sample units.

4.4.1.3.1 Sampling plan - Statistical sampling and inspection shall be in accordance with MIL-STD-105 for ordinary inspection. The acceptable quality levels (AQL) shall be as specified in Table III. Major and minor defects shall be as defined in MIL-STD-105.

TABLE III. Group A inspection

| Examination | Requirement paragraph | Method paragraph | AQL (% defective) | |
|---|---|---|---|---|
| | | | Major | Minor |
| Visual and mechanical examination | 3.2, 3.3, 3.7 and 3.8 | 4.5.1 | 1.5 | 6.5 |

4.4.1.4 Group B inspection - Group B inspection shall consist of the tests specified in Table IV, in the order shown.

TABLE IV. Group B inspection

| Test | Requirement paragraph | Method paragraph |
|---|---|---|
| Weight of Zinc coating | 3.4 | 4.5.2 |
| Adherence of Zinc coating | 3.5 | 4.5.3 |
| Gripping strength 1/ | 3.6 | 4.5.4 |

1/ If wire breakage occurs during this test the test is discontinued and shall be redone upon a new sample.



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4.4.1.4.1 Sampling plan - The sampling plan shall be in accordance with MIL-STD-105, for small-sample inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The AQL shall be 4.0 (percent defective), and the inspection level shall be S4 for normal and tightened inspection, and S3 for reduced inspection. For inspecting small lots with sample size code letters AA, A, and B, the acceptance number shall be zero.

4.4.1.4.2 Disposition of sample units - Sample units which have been subjected to group B inspection shall not be delivered on the contract or order.

4.4.1.4.3 Non-compliance - If a sample fails to pass group B inspection, the supplier shall take corrective action on the materials or process, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which were considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Group A inspection may be reinstituted; however, final acceptance shall be withheld until the group B reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and the corrective action taken shall be furnished to the contracting officer.

4.4.2 Inspection of preparation for delivery - Sample items and packs shall be selected and inspected to verify conformance with the requirements in section 5 of this specification.

4.5 Methods of examination and test -

4.5.1 Visual and mechanical examination - Single tube wire splicing sleeves shall be examined to determine that the material, design, construction, physical dimensions and workmanship are in accordance with the applicable requirements. (See 3.2, 3.3, 3.7 and 3.8)

4.5.2 Weight of Zinc coating (see 3.4) - Galvanized steel single tube wire splicing sleeves, only, shall be tested in accordance with the hydrochloric acid-antimony chloride method of A.S.T.M. Standard A90.

4.5.3 Adherence of Zinc coating (see 3.5) - Galvanized steel single tube wire splicing sleeves, only, shall be bent through an angle of ninety degrees around a cylindrical mandrel having a diameter equal to three times the nominal diameter of the single tube wire splicing sleeve under test. It shall then be examined for compliance with the applicable requirements.



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4.5.4 C[illegible] ... with sleeve compressed ... Tool TL-217. (Micropress No. [illegible] QC) as made by [illegible] Telephone Supply Co., or equal. The pressing shall be done using a tool properly adjusted in accordance with Table VI. Single tube wire splicing sleeves shall then be subjected to the tensile pulls of Table V.

TABLE V. Tensile pull—wire data

| Type | Wire | Tensile pull (in pounds) | Wire material | Size of Wire | | |
|---|---|---|---|---|---|---|
| | | | | Gage | Decimal equivalent (in inches) | Metric equivalent 2/ |
| C-080C | W-153 | 770 | Copper-clad steel | 12AWG | .080 | 2.03mm |
| C-104C | W-74 | 95% of average ascertained breaking strength 1/ | Hard drawn copper | 10AWG | .102 | 2.59mm |
| C-104Q | | 1172 | Copper-clad steel | 10AWG | .102 | 2.59mm |
| S-109Q | W-145 | 95% of average ascertained breaking strength | Galvanized steel | 12BWG | .109 | 2.77mm |
| S-134Q | W-155 | 95% of average ascertained breaking strength | Galvanized steel | 10BWG | .134 | 3.40mm |

1/ Five pieces of wire shall be tested to ascertain the average breaking strength.

2/ Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.



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TABLE VI. Pressing Data

| Type | Number of presses each end | Maximum diameter of Pressed Sleeve 1/ (in inches) |
|---|---|---|
| C-080C | 2 | 0.185 (4.70mm) 2/ |
| C-104C | 2 | 0.185 (4.70 mm) |
| C-104Q | 4 | 0.235 (5.97mm) |
| S-109Q | 4 | 0.235 (5.97mm) |
| S-134Q | 4 | 0.235 (5.97mm) |

1/ Minimum diameter is measured in a plane perpendicular to that of the fins formed by the pressing process.

2/ Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (see 6.2(c) ).

5.1.1 Level A

5.1.1.1 Cleaning - Single Tube Wire Splicing Sleeves shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying - Single Tube Wire Splicing Sleeves shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Unit packaging - Single Tube Wire Splicing Sleeves shall be packaged method III in accordance with MIL-P-116, 100 each per unit package, in a manner to insure compliance with the General (Method of Preservation) and Physical Protection paragraphs therein.



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5.1.2 Level C - Single Tube Wire Splicing Sleeves shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity. This package may conform to the suppliers commercial practice for retail distribution when it meets the requirements of this level.

5.2 Packing - Packing shall be level A, B or C as specified (see 6.2(c) ).

5.2.1 Level A - A quantity of single tube wire splicing sleeves packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, grade A, style E or MIL-B-43291. The gross weight of boxes conforming to PPP-B-640 shall not exceed 200 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97, type IV and PPP-S-760, type II respectively; Selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.2 Level B - Unless otherwise specified, a quantity of single tube wire splicing sleeves packaged as specified in 5.1, shall be packed as specified in 5.2.1 except that fiberboard boxes shall conform to PPP-B-640, class 1, grade A, style E or PPP-B-636, class domestic, variety DW, special requirements use. Containers conforming to PPP-B-640 shall be reinforced as specified in 5.2.1 except nonmetallic strapping conforming to PPP-S-760 shall be type 1.

5.2.3 Level C - A quantity of single tube wire splicing sleeves packaged as specified in 5.1, shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

5.3 Marking - In addition to any special marking required by the contract or order, each unit package, intermediate, and exterior container shall be marked in accordance with MIL-STD-129.

5.4 General - Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical items to the greatest extent possible.

6. NOTES

6.1 Intended use - Single tube wire splicing sleeves covered by this specification are intended for use in making butt splices in the solid, bare wires, used in telephone or open communication line construction.



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6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type(s) of single tube wire splicing sleeve (see Fig. 1).

(c) Levels of preservation, packaging and packing.

(d) When rough handling or functional test is required.

6.2.1 Indirect shipments - The preservation, packaging, packing, and marking specified in section 5 are intended for direct purchases by or direct shipments to the Government and are not intended to apply to contracts or orders between the supplier and the prime contractor.

6.3 Substitutions for specified products - If the bidder desires to substitute another material or fabricated part in those cases where the specification permits an "or equal" he shall submit a statement to that effect with his bid describing the proposed substitutions and shall submit data that will substantiate his claims that such substitutions are the equal of those specified. At the discretion of the contracting officer, samples may be required which will demonstrate by test the suitability of the proposed substitute.

Custodians:

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Preparing activity:

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Review activities:

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

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